

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08 Live sessions**90** min Per session**04** weeks Course duration**Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

**Built for beginners**

No prior Python or machine learning required

**Live code-along on Civil Engineering datasets**

Materials · Buildings & Energy · Transport · Environment · Geotechnics

**Hands-on notebooks, cross-domain assignments & solutions**

Plus a certificate and session recordings*

* Recordings shared after each session; available until two weeks after the course ends.

Course outline

01 Python for Civil Engineers
AI/ML basics, Colab, NumPy & pandas with engineering data

02 From Raw Data to ML-Ready
Explore, clean, visualize & split engineering data

03 Your First Prediction Model
Linear regression, error metrics & cross-validation

04 Trees & Ensemble Models
Decision Trees, Random Forest & boosting for nonlinear problems

05 Classification for Engineering Decisions
Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks
Architecture, activations, training & overfitting

07 Explainable AI & Engineering Judgement
Feature importance, SHAP, and engineering interpretation

08 From Prediction to Probability
Monte Carlo, ML surrogates & probability of failure

 **Designed for Civil Engineers across disciplines** — Materials, Structures, Geotechnics, Transportation, Water & Environment.

INSTRUCTOR

Dr. Abdul Waris

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

 Research Excellence Award 2026, IIT Hyderabad

 Invited talks at Oregon State University (USA) and a Faculty Development Programme at D.Y. Patil (Pune, India)

 Invited resource person, five-day ML workshop at NIT Srinagar

ONE ML WORKFLOW. MULTIPLE CIVIL ENGINEERING APPLICATIONS.

Problem → Data → Model → Validate → Interpret → Decide

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty
Practising engineers / Complete beginners to AI & ML

STARTS · LIMITED SEATS



**Saturday,
17 October 2026**

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